

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

Semester Final Examination

Course No.: EEE 4483

Course Title: Digital Electronics and Pulse Techniques

Summer Semester, A. Y. 2023-2024

Time: 3 Hours

Full Marks: 150

There are 6 (six) questions. Answer all 6 (six) questions. The symbols have their usual meanings. Marks of each question and corresponding COs are written in the brackets.

1. a) Using graphical analysis, explain how the operating point (Q-point) of a diode is determined in the exponential model. 10
(CO1)
- b) Answer the following questions: 6
(CO1)
- i) State the fundamental difference in the control mechanism of current in FETs versus BJTs.
 - ii) Explain the reason a BJT is termed a bipolar device, whereas an FET is termed a unipolar device.
 - iii) Compare the input impedance of a BJT with that of an FET, and explain how this affects circuit design.
- c) Explain the significance of the terms "Metal-Oxide" and "Field-Effect" in a MOSFET. Discuss the roles of the gate, oxide layer, and substrate in channel formation in a MOSFET, and how this leads to the high input impedance of the device. 9
(CO1)
2. a) The circuit in Fig. 2(a) shows a 3-input OR gate implemented using diode logic (DL). Each input (V_1, V_2, V_3) can take either 0V (Logic 0) or 5V (Logic 1). The diodes are assumed to have a forward voltage drop $\approx 0.7V$. For the circuit: 15
(CO2)
- i) Determine the output voltage V_{out} for all possible input conditions.
 - ii) From your results, state the corresponding logic output levels of the circuit.

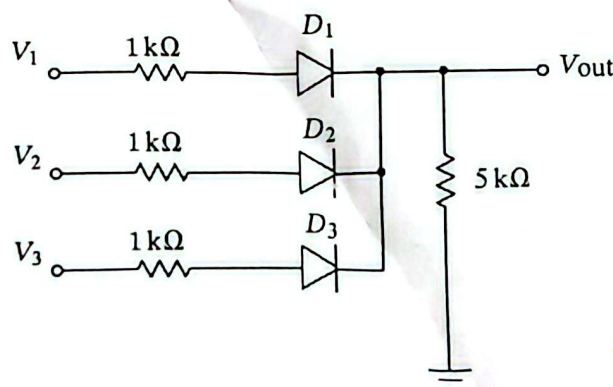


Fig. 2(a)

b) Define the following characteristic parameters of logic families:

10
(CO2)

- i) HIGH-level output voltage, V_{OH}
- ii) LOW-level output current, I_{OL}
- iii) Fall time, t_f
- iv) Propagation delay, t_p
- v) Fan-in

3.

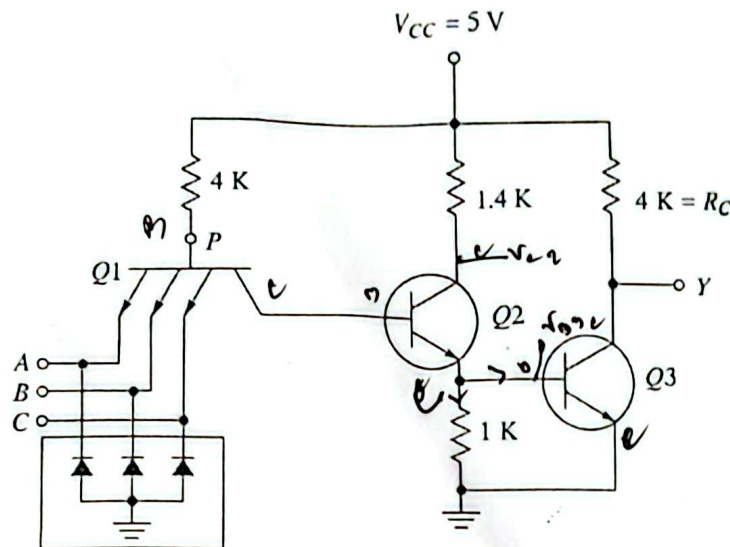


Fig. 3

Using Fig. 3, answer the questions 3(a) to 3(c) –

- a) If all the inputs are HIGH in Fig. 3, then determine the emitter current of Q_3 . You can assume following values for the transistors: $V_{BE}(\text{sat}) = 0.8 \text{ V}$, $V_{CE}(\text{sat}) = 0.2 \text{ V}$, $V_{BC}(\text{on}) = 0.7 \text{ V}$. 7
(CO2)
- b) Explain the purpose of the three diodes connected to inputs A, B, and C in the circuit of Fig. 3. 3
(CO2)
- c) The circuit shown in Fig. 3 suffers from delays in switching speed due to load capacitance at the output node Y. 15
(CO2)
 - i) Define load capacitance in logic circuits and explain how it affects the switching speed in the circuit of Fig. 3.
 - ii) Illustrate an improved version of the circuit in Fig. 3 that minimizes the effect of load capacitance.
 - iii) Explain how your modification minimizes the effect of load capacitance and improves the switching performance.

4. a) The circuit shown in Fig. 4(a) is a primitive theoretical model of a Sample-and-Hold (S/H) circuit. Using this figure, answer the questions (i) to (iii) –

13
(CO3)

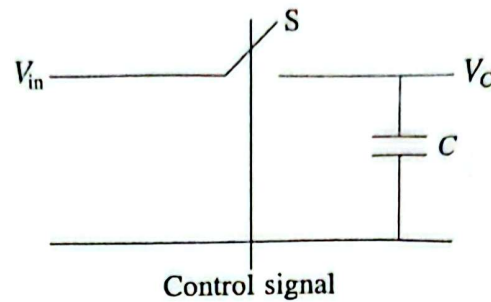


Fig. 4(a)

- i) Explain the necessity of a Sample-and-Hold circuit in an Analog-to-Digital Converter (ADC) system.
 - ii) Identify at least three major practical issues that arise if this circuit is directly implemented as shown in Fig. 4(a).
 - iii) Propose and illustrate an improved practical S/H circuit to overcome the limitations identified in part (ii). Clearly explain the role of each added component.
- b) A single-slope ADC integrates a reference voltage (using ramp generator) until it matches the input analog voltage. Suppose the clock frequency is 50 kHz, the counter is 8-bit, and the reference ramp has a slope of 1.953 V/ms.
- (i) Find the quantization step size of the ADC in volts. (The quantization step size is the smallest input change that results in one count)
- (ii) Calculate the digital output for an input voltage of 3.91 V.
- (iii) Determine the maximum conversion time (The time an ADC takes for conversion when the input is full-scale, i.e., when the counter reaches its maximum count)
5. a) A 4-bit Successive Approximation ADC operates with a reference voltage of 8 V. For an input voltage of 5.3 V, determine the final 4-bit digital output code in binary form. Show, step by step, how the SAR determines each bit of the digital output from MSB to LSB.
- b) Consider a dual-slope ADC, where the input voltage is integrated (run-up) for a fixed time of 40.96 ms. The clock frequency is 100 kHz, and the reference voltage is 5 V. For an input of 2.41 V:
- (i) Calculate the number of counts accumulated during the integration phase (run-up).
 - (ii) Determine the number of bits in the counter.
 - (iii) Find the de-integration time required for the reference to bring the ramp output back to zero.
 - (iv) Determine the digital output code (decimal and binary) corresponding to the 2.41 V input.

6. a) With proper illustrations, explain the working principle of single slope ADC. Discuss the limitations of a single-slope ADC and explain how a dual-slope ADC overcomes these limitations. 18
(CO3)
- b) Define the following response parameters of Sample-and-Hold circuit: 7
(CO3)
- i) Aperture time
 - ii) Acquisition time
 - iii) Droop rate

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